Unit 1, Lesson 2: Evaluating Functions in Function Notation

Benchmark

MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.

Learning Target

- [ ] I can evaluate a function for a specific input value of the domain.

1.2.1 Warm-Up: Food and Research Career

Richard works in Research and Development for Kellogg's. One of the projects he is most proud of involves developing a line of organics products and no added sugar products. Richard also points out that working with a team provides an exciting variety of experiences that is crucial and rewarding to the work of the company.

1. How could mathematics help in the food science industry?

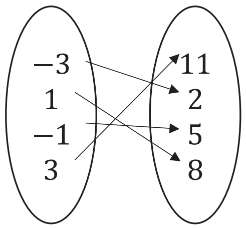
2. Why is it important to have teamwork skills when working for a company?

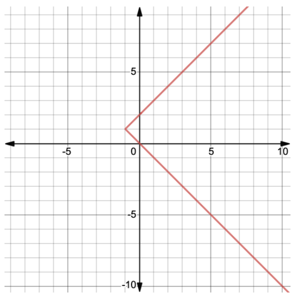
1.2.2 Collaboration: Functions

A ***relation*** is a set of input-output pairs.

A ***function*** is a mathematical relation for which each element in the domain corresponds to exactly one element in the range.

Three different relations are shown, with each represented in a different way.





x	y
-4	2
7	0
-4	-4
0	-6

1. With your partner, write an explanation identifying which of the relations above are functions.

1.2.3 Exploration: Graphs of Functions

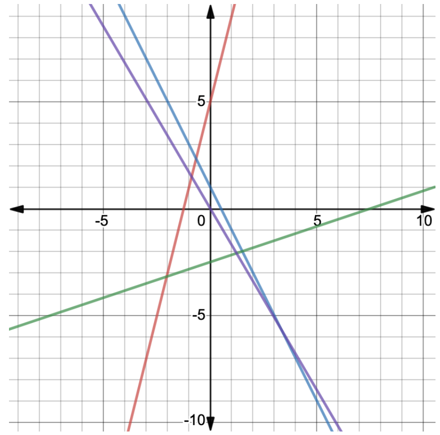
The equations and graphs of four linear functions are shown.

$y = 4x + 5$

$y = -2x + 1$

$y = \frac{1}{3}x - 2.5$

$y = -1.7x$



1. Write the equations of the lines that decrease from left to right.

2. Write the equations of the lines that increase from left to right.

3. What do the equations of the lines that increase from left to right have in common?

4. How are the equations of the lines that decrease from left to right different from the equations of the lines that increase from left to right?

1.2.4 Guided Instruction: Evaluating Functions

Each of the functions in the Exploration starts with "$y =$." If asked to use the function y from the list of functions to find a specific value, it would not be clear which of the four functions to use.

Since mathematicians often work with multiple functions at a time, ***function notation*** is used to name specific functions.

The notation $y = f(x)$ defines a function named f. This indicates "f is a function of x." The letter x represents the input value, or independent variable. The letter y, or $f(x)$, represents the output value, or dependent variable. The notation $f(x)$ is read, "f of x."

name of function　　　　input of function

$$\underbrace{f(x)}_{\text{output of function}}$$

A function can be named using any letter. For example, the previous functions can be renamed as shown:

$f(x) = 4x + 5$　　　　$g(x) = -2x + 1$

$h(x) = \frac{1}{3}x - 2.5$　　　　$c(x) = -1.7x$

Now, if asked to use the function h to find a specified value of x, it would be clear which function to use.

Functions are evaluated in the same way equations are evaluated when given a value for a variable.

Consider the function $f(w) = -7w^2 - 10w + 3$.

1. Find the value $f(10)$.

$$f(w) = -7w^2 - 10w + 3$$

$$f(10) = -7(10)^2 - 10(10) + 3$$

$$f(10) = \underline{\qquad\qquad}$$

When the input is 10, the output of the function is _____.

Using function notation, this is written $f(10) =$ _____.

2. Find the value $f(-3)$.

$$f(-3) = \underline{\qquad\qquad}$$

This same function $f(w)$ can be represented in other ways.

Shown here are a table of values, a graph, and a mapping diagram. The function can be evaluated from each of these representations.

Table of Values

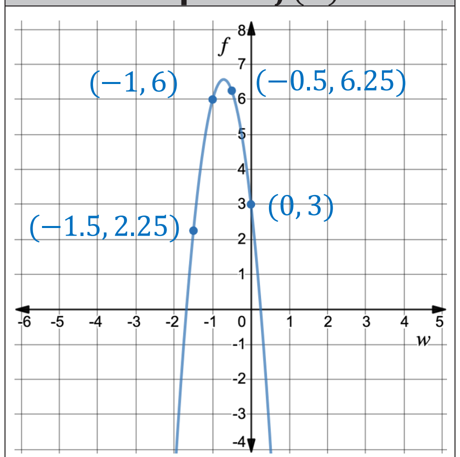
w	$f(w)$
-15	-1422
-10	-597
-1	6
2	-45

3. Find $f(-10)$.

   $f(-10) =$ _____

4. If $f(w) = -45$, what is the value of w?

Graph of $f(w)$

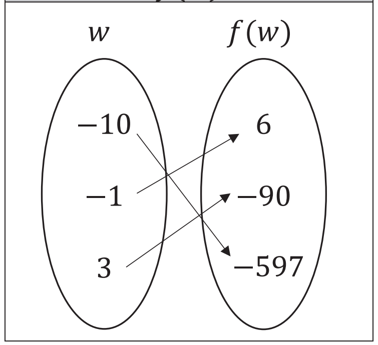


Find each value from the graph.

5. $f(-1.5) =$ _____

6. $f(0) =$ _____

Mapping Diagram of $f(w)$



Find each value from the mapping diagram.

7. $f(3) =$ _____

8. Find the value of w when $f(w) = 6$.

1.2.5 Try It: Function Notation

1. Given the function $c(y) = -6y - 5$, find each value.

   a. $c(-4) =$

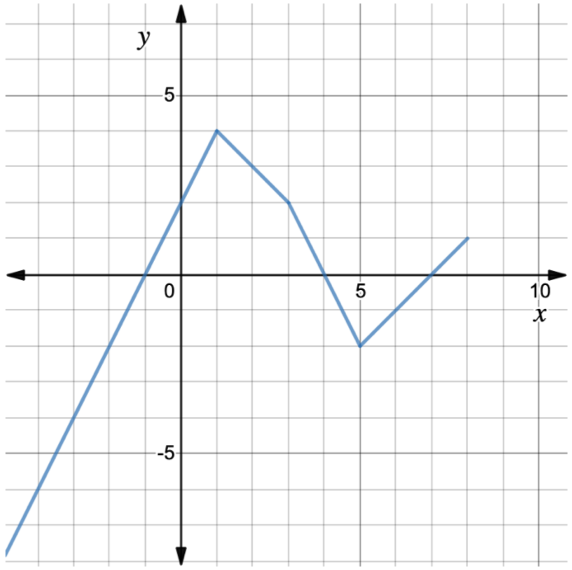
   b. $c\left(\frac{3}{4}\right) =$

   c. For what value of y is $c(y) = 9$?

2. Given $g(k) = 2k^2 - 4$, find $g(-5)$.

3. Given $h(x) = \sqrt{2.5x + 1}$, find $h(3.2)$.

4. Use the graph of $y = f(x)$ below to find each value.



   a. $f(1) =$

   b. $f(-2) =$

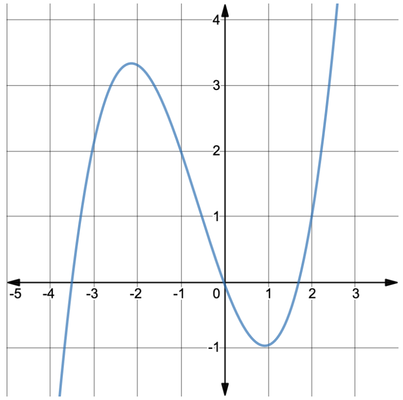
   c. For what value of x is $f(x) = -4$?

1.2.6 Your Turn!

1. Find each value given the function $m(t) = \frac{t}{3} + 6$.
   a. $m(-9) =$

   b. Find the input for which $m(t) = 13$.

2. Find each value using the graph of the function $f(x)$.



   a. $f(-3.5) =$

   b. $f(2) =$

   c. Find x when $f(x) = 4$.

3. Use the table of values representing $D(r)$ to find each value.
   a. $D(-4) =$

   b. $D(-6) =$

r	$D(r)$
-6	9
0	-5
-4	9
2	0
9	7

   c. Is $D(r)$ a function? Explain why or why not.

1.2.7 Wrap-Up: Ticket Out the Door

1. Given the function $h(t) = t^3 - 2t$, find $h(-7)$.

2. Use the graph of $q(x)$ to find $q(-3)$.